

**Desing and Development of a Web-Based Mamagan Fun & Adventure Beach Resort
Online Booking and Payment System with GCash Integration using PayMongo**

A System Integration and Architecture Project
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Fiel, Jerome Kevin D.

Mr. Jeremiah Dela Cruz
Adviser

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INTRODUCTION

The advancement of technology has greatly influenced the way businesses manage their daily operations, especially in the tourism and hospitality industry. Beach resorts, hotels, and recreational establishments are now encouraged to adopt digital solutions to improve customer service, organize transactions, and provide faster access to information. In many resorts, however, booking and reservation processes are still handled manually through walk-in inquiries, phone calls, handwritten logbooks, or social media messages. While these methods may still work, they often lead to problems such as delayed responses, misplaced records, double bookings, inaccurate availability updates, and difficulty in monitoring payments and guest schedules.

The Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration using PayMongo was developed to provide a modern, organized, and convenient solution for managing resort reservations. This system allows guests to browse available resort facilities, view rates and descriptions, check availability, create bookings, and pay online using supported digital payment methods. Through the integration of PayMongo, the system can support online payments such as GCash, Maya, GrabPay, and card transactions. After successful payment, the system can generate a digital QR ticket that serves as the guest's proof of reservation and can be used during check-in at the resort.

The main purpose of this project is to improve the booking process of Mamagan Fun & Adventure Beach Resort by reducing manual work and providing a more efficient reservation system for both guests and resort staff. For guests, the system offers convenience because they can make reservations anytime and anywhere without needing to visit the resort physically or wait for manual confirmation. For resort staff, the system provides an admin dashboard where they can manage facilities, update rates and promos, monitor bookings, block unavailable dates, confirm guest arrivals, and generate reports. This helps improve the accuracy of records and makes daily resort operations more organized.

This project is important because it addresses common challenges experienced in manual booking systems. Manual reservation methods can cause confusion, especially during peak seasons when many customers inquire or book at the same time. Without a centralized system, resort staff may find it difficult to track available facilities, payment status, guest information, and scheduled reservations. By using an online booking and payment system, the

resort can minimize errors, prevent overbooking, improve transaction monitoring, and provide faster confirmation to customers.

Furthermore, the system supports the growing need for cashless and contactless transactions. Since many customers now prefer digital payment methods such as GCash and other online payment platforms, integrating a payment gateway into the system makes the booking process more convenient and secure. It also helps the resort ensure that reservations are properly paid before confirmation, reducing the chances of unpaid or unconfirmed bookings.

Overall, the development of the Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration using Paymongo aims to provide a reliable, user-friendly, and efficient platform for resort reservation management. It is designed to improve customer experience, assist resort staff in managing operations, and support the resort's transition from manual booking procedures to a more modern and technology-based system.

Background of the Study

The advancement of technology has changed the way businesses provide services and manage daily operations. In the tourism and hospitality industry, many resorts are now encouraged to use digital systems to improve customer service, organize reservations, and make transactions faster and more reliable. However, some beach resorts still depend on traditional booking methods such as walk-in reservations, phone calls, handwritten logbooks, and social media messages. These manual processes may be difficult to manage, especially during peak seasons when many guests inquire, reserve, or cancel bookings at the same time.

Mamagan Fun & Adventure Beach Resort offers different facilities and activities for guests, such as cabanas, day beds, jet skis, and island tours. Because of this, the resort needs an organized system that can manage reservations, facility availability, payments, and guest confirmation efficiently. Without a centralized system, staff may experience problems such as misplaced records, double bookings, delayed confirmation, and difficulty in monitoring payments. To address these concerns, the Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration was developed as a web-based platform that supports online reservations, digital payments, and QR ticket generation.

Purpose of the System

The purpose of the system is to provide a convenient, organized, and efficient online booking platform for Mamagan Fun & Adventure Beach Resort. The system allows guests to browse resort facilities, check available dates and time slots, create reservations, and pay online using supported digital payment methods such as GCash, Maya, GrabPay, and card payments through PayMongo. After successful payment, the system generates a digital QR ticket that serves as proof of reservation and can be used during guest check-in.

For resort staff, the system provides an admin dashboard where they can manage facilities, update rates and promos, block unavailable dates, monitor bookings, confirm guest arrivals, and generate reports. Through this system, the resort can reduce manual work, improve record accuracy, and provide faster service to guests.

Importance of the Project

This project is important because it helps improve both customer experience and resort management. For guests, the system provides convenience because they can make reservations anytime and anywhere without needing to visit the resort physically or wait for manual confirmation. It also allows them to complete payments through online payment methods, making the transaction faster and more accessible.

For resort staff, the system helps organize booking records, payment details, facility schedules, and guest information in one platform. This reduces the chance of human error and allows staff to manage daily operations more effectively. The system also supports cashless transactions, which are becoming more common and preferred by many customers. Overall, the project contributes to a more modern, reliable, and professional booking process for the resort.

Problems the System Aims to Solve

The system aims to solve common problems found in manual reservation and payment processes. One of the main problems is overbooking, which can happen when facility availability is not updated properly. Another problem is delayed confirmation, especially when staff manually respond to inquiries and payment updates. Manual recordkeeping may also result in lost, duplicated, or inaccurate booking information.

The system also addresses the difficulty of tracking payment status. In a manual setup, staff may need to check payment screenshots or messages one by one, which can be time-consuming and prone to mistakes. Through payment gateway integration, the system can process and confirm online payments more efficiently. In addition, the QR ticket feature helps make guest verification easier during arrival.

OBJECTIVE OF THE PROJECT

General Objective

The general objective of this project is to design and develop a Web-based online booking and payment system for Mamagan Fun & Adventure Beach Resort that will provide guests with a convenient way to reserve resort facilities, complete online payments through GCash and other supported digital payment methods and receive digital booking confirmation through a QR ticket. The system also aims to help resort staff manage reservations, facilities, payments, guest check-ins, and reports in a more organized and efficient manner.

Specific Objectives

1. To develop a user-friendly online booking system that allows guests to browse resort facilities, view rates, and check available schedules.
2. To provide a secure registration and login system for guests and resort staff.
3. To allow guests to create and manage reservations online without relying on manual booking methods.
4. To integrate online payment through PayMongo, supporting GCash, Maya, GrabPay, and card payments.
5. To generate a digital QR ticket after successful payment as proof of reservation and for faster guest check-in.
6. To provide an admin dashboard where resort staff can manage facilities, bookings, rates, blocked dates, payments, and reports.

SCOPE AND LIMITATIONS OF THE PROJECT

The Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration is a web-based system designed to make resort reservations more convenient and organized. It allows guests to create an account, log in, browse available facilities, check schedules, make bookings, and pay online using GCash, Maya, GrabPay, or card payments through PayMongo.

After successful payment, the system generates a digital QR ticket that serves as proof of reservation and can be used during guest check-in. For the admin side, resort staff can manage facilities, rates, blocked dates, bookings, payments, guest arrivals, and reports through the admin dashboard.

Overall, the system helps reduce manual booking errors, prevent overbooking, improve payment monitoring, and provide a faster and more organized reservation process for both guests and resort staff.

What the System Can Do

The Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration is designed to provide an online platform for managing resort reservations and payments. The system allows guests to create an account, log in, browse available resort facilities, check schedules, make reservations, and pay online using supported digital payment methods. After successful payment, the system can generate a digital QR ticket that serves as proof of reservation and can be used during guest check-in.

The system also allows resort staff or administrators to manage resort operations through an admin dashboard. Admin users can manage facilities, update rates, monitor bookings, verify payments, block unavailable dates, manage promotions, confirm guest arrivals, and view reports. Through these functions, the system helps reduce manual booking errors, prevent overbooking, and improve the organization of resort records.

Features Included in the System

The project includes the following features:

1. Guest registration and login system.
2. Guest profile and booking management.
3. Facility browsing with descriptions, rates, and availability.
4. Online booking and reservation creation.
5. Real-time availability checking by date and time slot.
6. Online payment integration using PayMongo.
7. Support for GCash, Maya, GrabPay, and card payments.
8. Automatic payment confirmation through the payment gateway.
9. Digital QR ticket generation after successful payment.
10. Admin dashboard for resort staff.
11. Facility, rate, promo, and blocked date management.
12. Booking monitoring and guest check-in confirmation.
13. Revenue and occupancy reporting.
14. Centralized database for users, facilities, bookings, payments, and QR tickets.

Limitations of the Project

Although the system provides an organized and efficient booking process, it also has limitations. First, the system requires an internet connection because it is web-based. Guests and resort staff must be connected to the internet to access the website, process bookings, confirm payments, and manage reservations.

Second, the online payment feature depends on the availability and service performance of PayMongo. If the payment gateway experiences downtime, delayed confirmation, or technical issues, the system may not be able to process or confirm payments immediately. The system can support online payment methods such as GCash, Maya, GrabPay, and cards only through the payment gateway.

Third, the system is limited to the facilities and services registered by the resort admin. It can only display and manage facilities that are already encoded in the database. If the resort adds new services or activities, the admin must manually add them to the system before guests can view or book them.

Fourth, the system does not fully automate refund processing. Refunds may still depend on the resort's cancellation and refund policy. If a guest cancels a booking, the admin may need to review the cancellation first before approving any refund request.

Fifth, the system mainly focuses on online booking and payment transactions. Walk-in guests may still need to be handled manually by resort staff unless their bookings are encoded into the system by the admin. The system does not automatically detect or process walk-in transactions unless the staff updates the records.

Sixth, the system's QR ticket verification depends on the availability of the admin dashboard or QR checking feature. If there is no internet connection at the resort during check-in, staff may need to verify the guest's reservation manually using saved records or booking details.

Lastly, the system does not include a mobile application version. It is designed as a web-based system that can be accessed through a browser. Although it may be accessible on mobile devices if properly designed, it is not a separate Android or iOS application.

SYSTEM DESCRIPTION

The Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration is a web-based system designed to improve the reservation and payment process of Mamagan Fun & Adventure Beach Resort. The system provides an online platform where guests can browse resort facilities, check available schedules, create bookings, pay online, and receive digital confirmation through a QR ticket. It helps replace manual booking methods such as handwritten logbooks, walk-in reservations, and message-based inquiries with a more organized and efficient digital process.

The system is developed to serve both guests and resort staff. Guests can easily access the website to view available facilities such as cottages, cabanas, day beds, jet skis, island tours, and other resort services. They can select their preferred date and time, submit a booking request, and proceed with payment using supported online payment methods. After payment confirmation, the system generates a digital QR ticket that serves as proof of reservation and can be presented during check-in at the resort.

For resort staff or administrators, the system provides an admin dashboard that allows them to manage facilities, booking records, payment status, promotions, blocked dates, seasonal rates, guest check-ins, and reports. This helps the resort maintain accurate records, prevent overbooking, monitor payments, and provide faster service to customers.

Overall, the system aims to make resort booking more convenient for guests and more manageable for resort personnel. It supports digital transformation by providing a centralized system for reservations, payments, customer records, and operational monitoring.

Major Modules and Features

1. User Registration and Login Module - This module allows guests, staff, and administrators to create and access their accounts securely. Guests can register by providing their personal details such as name, email, phone number, and password. The login system helps protect user information and ensures that only authorized users can access specific system functions.

Through this module, users can manage their accounts and view their booking records. Admin and staff accounts have additional access to management features depending on their role.

2. Facility Browsing Module - The facility browsing module allows guests to view the available resort facilities and services. These may include cottages, cabanas, day beds, jet skis, island tours, and other bookable resort activities. Each facility may display important details such as name, category, description, image, price range, capacity, rental type, and availability status.

This feature helps guests compare available options before making a reservation. It also provides clear information about the resort's services, making the booking process easier and more convenient.

3. Availability Checking Module - The availability checking module allows guests to check whether a facility is available on their selected date and time. This feature helps prevent scheduling conflicts and overbooking. The system checks existing bookings, blocked dates, inventory count, and other restrictions before allowing the guest to continue with the reservation.

4. Online Booking Module - The online booking module allows guests to create reservations through the website. Guests can select a facility, choose a booking date, start time, end time, number of guests, quantity, and booking type. The system calculates the base amount, discount amount, and total amount based on the selected facility, seasonal rate, or promotion.

After the booking is submitted, the system stores the reservation details in the database. The booking may have statuses such as pending, confirmed, cancelled, or completed depending on the payment and admin confirmation.

5. Online Payment Module - The payment module allows guests to pay for their reservations using supported digital payment methods. The system is designed to support PayMongo payment processing, which may include GCash, Maya, GrabPay, and card payments. Payment

details such as amount, method, reference number, payment status, checkout ID, payment ID, and proof of payment are stored in the system.

6. QR Ticket Generation Modul - After successful payment confirmation, the system generates a digital QR ticket for the guest. This ticket contains booking information and serves as proof of reservation. The guest can present the QR ticket upon arrival at the resort.

For staff, the QR ticket makes the check-in process faster and more reliable. Instead of manually searching through booking records, staff can use the QR ticket details to verify the guest's reservation and confirm arrival.

7. Admin Dashboard Module - The admin dashboard is used by resort staff or administrators to manage the overall system. Through this module, admins can view bookings, manage facilities, update facility details, monitor payment status, confirm check-ins, and generate reports.

8. Facility Management Module - The facility management module allows admins to add, edit, update, activate, deactivate, or remove resort facilities. Admins can manage facility names, categories, descriptions, images, prices, capacity, inventory count, rental type, and bookable status.

9. Promotions and Seasonal Rates Module - This module allows the admin to create and manage promotional discounts and seasonal pricing. Promotions may include discount codes, discount types, discount values, minimum amount requirements, usage limits, start dates, and end dates. Seasonal rates can also be applied during specific dates, such as peak seasons, holidays, or special resort events. This feature helps the resort adjust pricing based on demand and marketing strategies.

10. Blackout Dates Module - The blackout dates module allows admins to block specific dates or times when a facility is not available. This may be used for maintenance, private events, full bookings, or temporary closure. This module helps prevent guests from booking unavailable facilities and supports better scheduling control.

11. Reviews and Ratings Module - The reviews and ratings module allow guests to leave feedback about resort facilities after using the service. Guests may provide ratings, comments, and uploaded media related to their experience.

This feature helps the resort collect customer feedback and improve service quality.

12. Reports and Monitoring Module - The reports module helps the admin monitor resort performance. It may include booking summaries, revenue records, occupancy reports, payment records, and guest activity logs.

User Roles and Functionalities

1. Guest / Customer

The guest is the main user of the system. Guests use the website to browse facilities, create bookings, pay online, and receive booking confirmation.

Guest Functionalities:

- Register and log in to an account.
- View available resort facilities and services.
- Check facility descriptions, rates, capacity, and availability.
- Select booking date, time, quantity, and number of guests.
- Create and manage reservations.
- Apply promotion codes, if available.
- Pay online using supported payment methods.
- Receive a digital QR ticket after successful payment.
- View booking history and payment status.
- Submit reviews and ratings for facilities.

2. Resort Staff / Admin

The resort staff or admin manages the operational side of the system. This user role has access to the admin dashboard and is responsible for monitoring bookings, facilities, payments, and guest check-ins.

Admin Functionalities:

- Log in to the admin dashboard.
- Add, edit, or update facility information.

- Manage facility rates, descriptions, images, and availability.
- View and monitor guest bookings.
- Confirm or update booking status.
- Check and verify payment status.
- Manage promotions and discount codes.
- Manage seasonal rates.
- Block unavailable dates for maintenance or events.
- Confirm guest arrival using QR ticket details.
- View reports such as revenue, occupancy, and booking summaries.
- Monitor system activity logs.

3. System Administrator

The system administrator is responsible for maintaining the system settings, user accounts, and overall system configuration. This role may have higher access than regular resort staff.

System Administrator Functionalities:

- Manage user roles and account status.
- Configure system settings.
- Monitor system logs.
- Maintain database records.
- Ensure that the system is functioning properly.
- Assist in troubleshooting technical issues.

4. Payment Gateway / PayMongo

PayMongo serves as the external payment gateway used by the system to process online payments. It is not a direct user but functions as an external system connected to the booking platform.

Payment Gateway Functionalities:

- Process online payments.
- Support GCash, Maya, GrabPay, and card transactions.
- Provide checkout links or payment references.
- Send payment confirmation details to the system.
- Help update payment status after successful or failed transactions.

TECHNOLOGIES TO BE USED

The Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration was developed using PHP, Laravel, MySQL, PayMongo, and GitHub. These technologies were used to create a functional web-based system for online booking, payment processing, database management, and project development.

Programming Language(s)

PHP - was used as the main programming language of the system. It handles the backend processes such as user registration, login, facility management, booking transactions, payment records, and admin functions.

Frameworks/Libraries

Laravel - was used as the main PHP framework of the system. It provides an organized structure for routes, controllers, models, middleware, authentication, and database management. Laravel also helps make the system easier to maintain, secure, and scalable.

Laravel Blade - was used as the templating engine for creating dynamic web pages. It allows the system to display data from the database, such as facilities, bookings, payments, and user information.

PayMongoAPI - was used for online payment integration. It allows the system to process digital payments through GCash and other supported payment methods.

Database Management System

MySQL - was used as the database management system of the project. It stores important records such as user accounts, resort facilities, bookings, payments, QR tickets, promotions, blocked dates, reviews, and system settings.

phpMyAdmin - was used to manage the MySQL database. It allows the developer to create tables, edit records, import and export SQL files, and monitor the database through a web-based interface.

Development Tools Used

Visual Studio Code - was used as the main code editor for writing and editing the system files, including PHP, Laravel, HTML, CSS, JavaScript, and configuration files.

XAMPP - was used as the local development environment. It provides Apache, MySQL, PHP, and phpMyAdmin, which are needed to run and test the system locally.

Composer - was used to install and manage Laravel packages and dependencies required by the system.

GitHub - was used for source code storage, backup, and version control. It helped track changes during development and organize the project files.

Web Browser - such as Google Chrome, was used to test the system's interface, features, and overall functionality.

SYSTEM DESIGN

This part explains how your system is designed using diagrams, database relationships, architecture, and interface screenshots. For your project, the **System Design** section should be based on your actual database setup in phpMyAdmin, especially the tables such as users, facilities, bookings, payments, tickets, promotions, blackout_dates, seasonal_rates, facility_reviews, app_settings, and system_logs.

Use case diagram

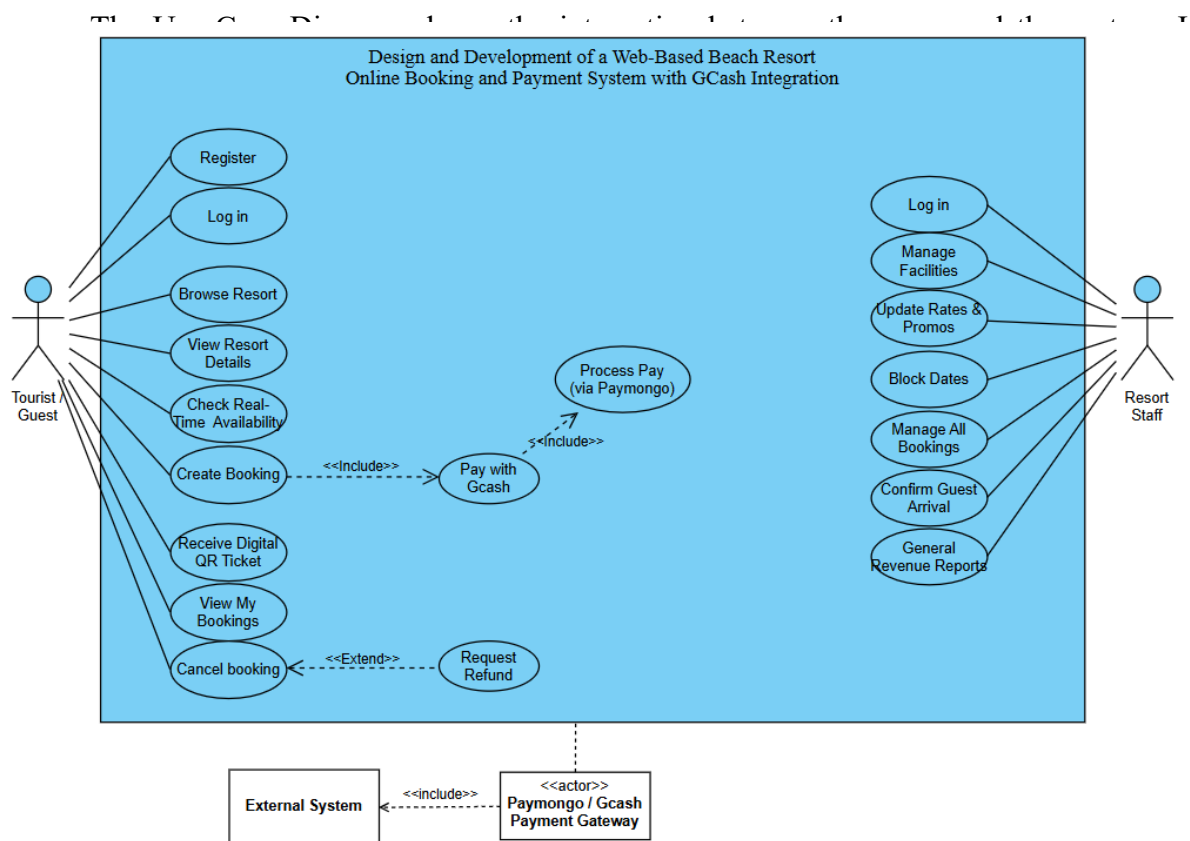


Figure 1: Use Case Diagram

Flowchart

The flowchart shows the step-by-step process of how the Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration works. It visually presents the flow of actions from guest registration, facility selection, booking creation, payment processing, QR ticket generation, and guest check-in confirmation. It also shows the admin p

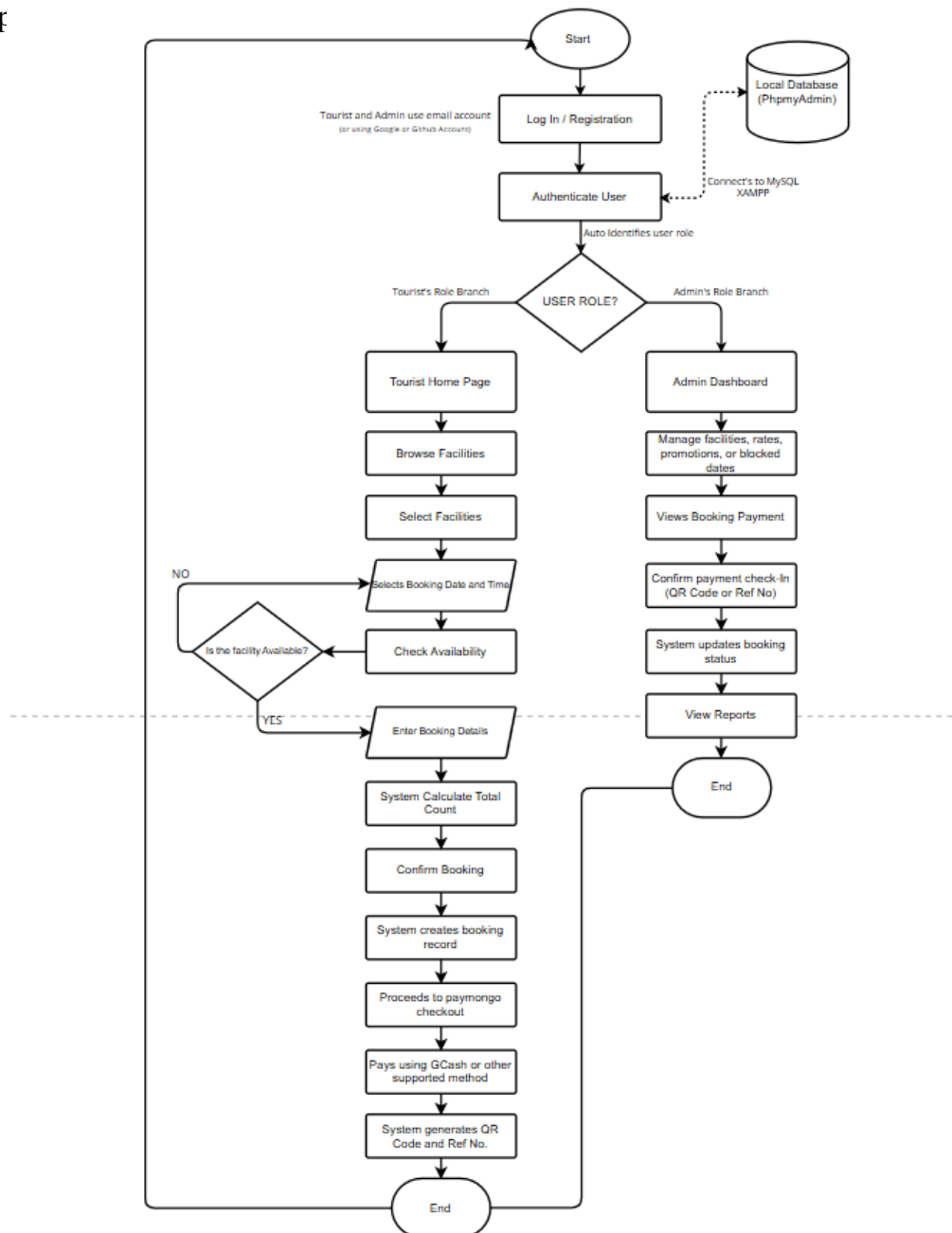


Figure 2: System Flowchart

Er diagram / database design

The ER Diagram, or Entity Relationship Diagram, shows the structure of the database used in the Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration. It presents the main tables, important fields, primary keys, foreign keys, and relationships between tables. This diagram helps explain how guest accounts, facilities, bookings, payments, QR tickets, promotions, reviews, and system records are connected in the database.

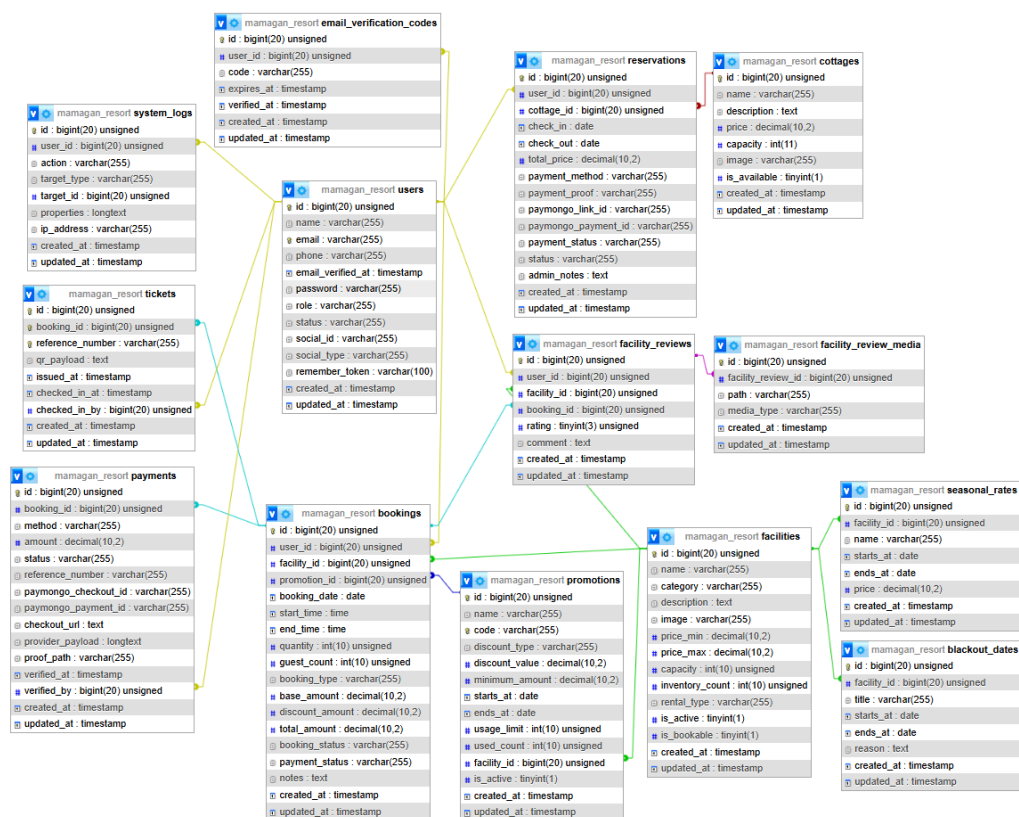


Figure 3: Database Design

System architecture

The System Architecture Diagram shows how the different parts of the system work together. It explains how guests and admins access the system, how Laravel processes requests, how MySQL stores data, and how PayMongo handles online payment transactions.

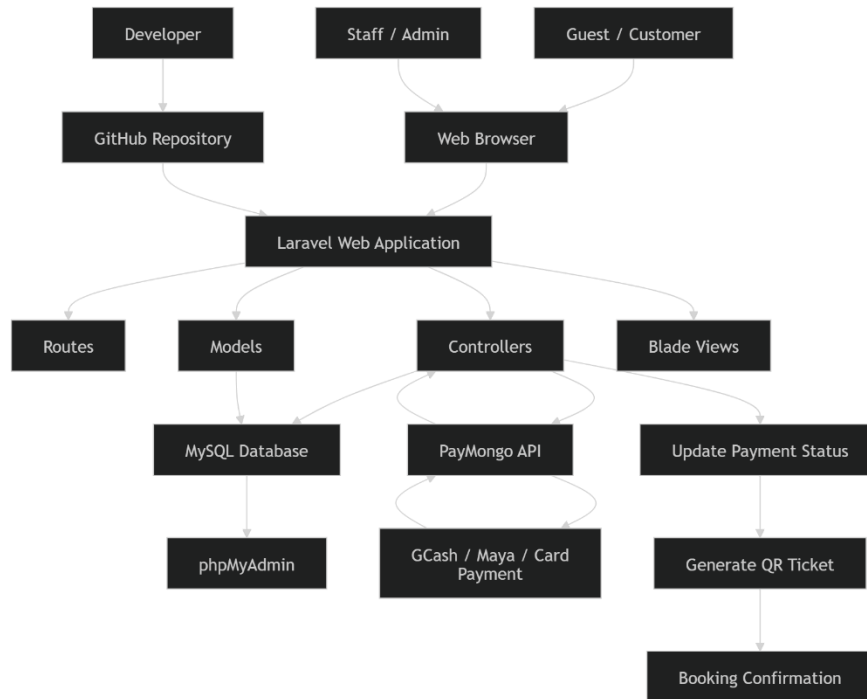


Figure 4: *System Architecture*

System Interface Design

The Interface Design section shows the actual pages and screens of the Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration. It helps demonstrate how users interact with the system, including guest booking, online payment, QR ticket generation, and admin management.

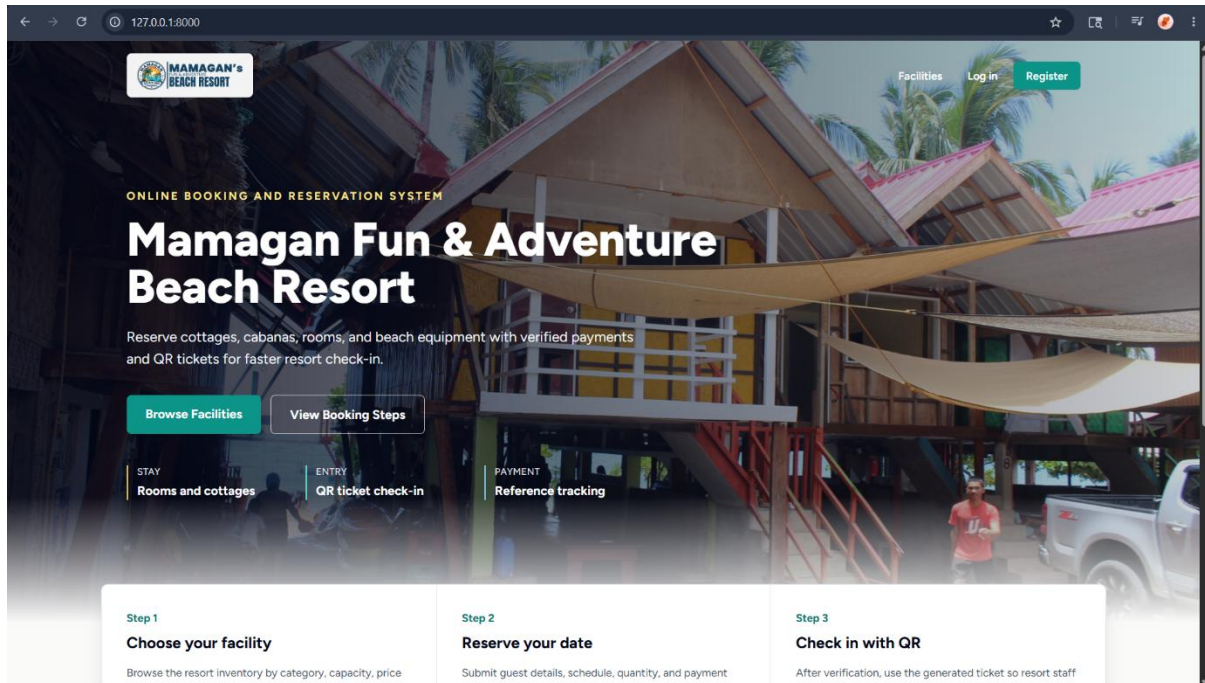
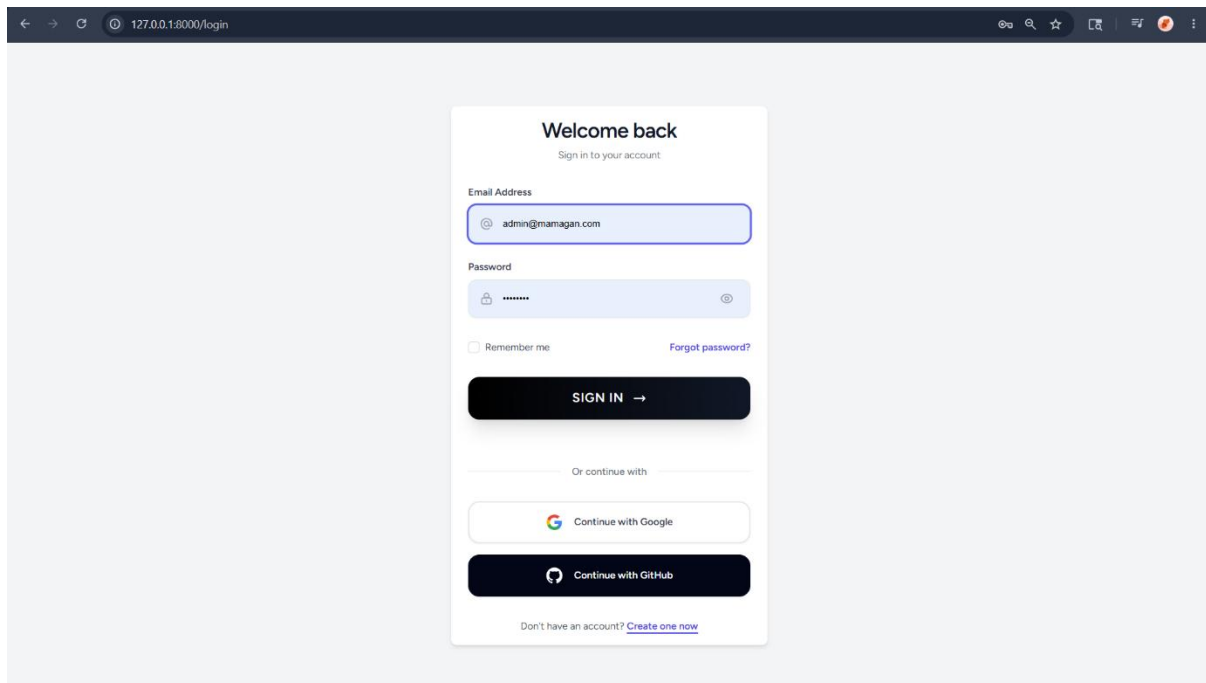


Figure 5: *Homepage of Mamagan Fun & Adventure Beach Resort Online Booking System*

The homepage serves as the main page of the system where guests can view general information about Mamagan Fun & Adventure Beach Resort. It provides navigation links to facilities, booking, login, registration, and other important pages.

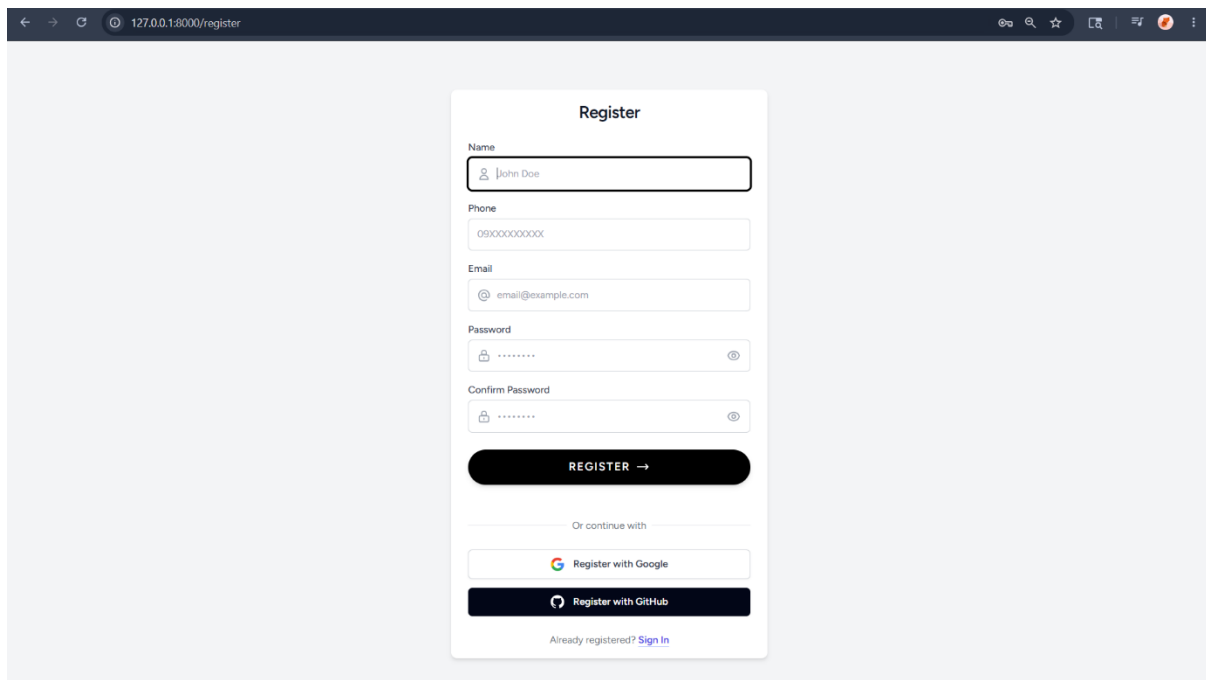


The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/login". The login form is titled "Welcome back" and "Sign in to your account". It contains the following elements:

- Email Address:** A text input field containing "admin@mamagan.com".
- Password:** A password input field with masked characters "*****".
- Remember me:** An unchecked checkbox.
- Forgot password?:** A link in purple text.
- SIGN IN →:** A large black button with white text.
- Or continue with:** A separator line with the text "Or continue with".
- Continue with Google:** A button with the Google logo.
- Continue with GitHub:** A button with the GitHub logo.
- Don't have an account? Create one now:** A link in purple text at the bottom.

Figure 6: *Login Interface*

The login page allows users to securely access their accounts. Guests can log in to manage bookings, while staff and admins can access management features depending on their assigned role.



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/register". The registration form is titled "Register" and contains the following elements:

- Name:** A text input field containing "John Doe".
- Phone:** A text input field containing "0900000000X".
- Email:** A text input field containing "email@example.com".
- Password:** A password input field with masked characters "*****".
- Confirm Password:** A password input field with masked characters "*****".
- REGISTER →:** A large black button with white text.
- Or continue with:** A separator line with the text "Or continue with".
- Register with Google:** A button with the Google logo.
- Register with GitHub:** A button with the GitHub logo.
- Already registered? Sign In:** A link in purple text at the bottom.

Figure 7: *Guest Registration Interface*

The registration page allows new guests to create an account before making a reservation. The information entered by the guest is stored in the users table of the database.

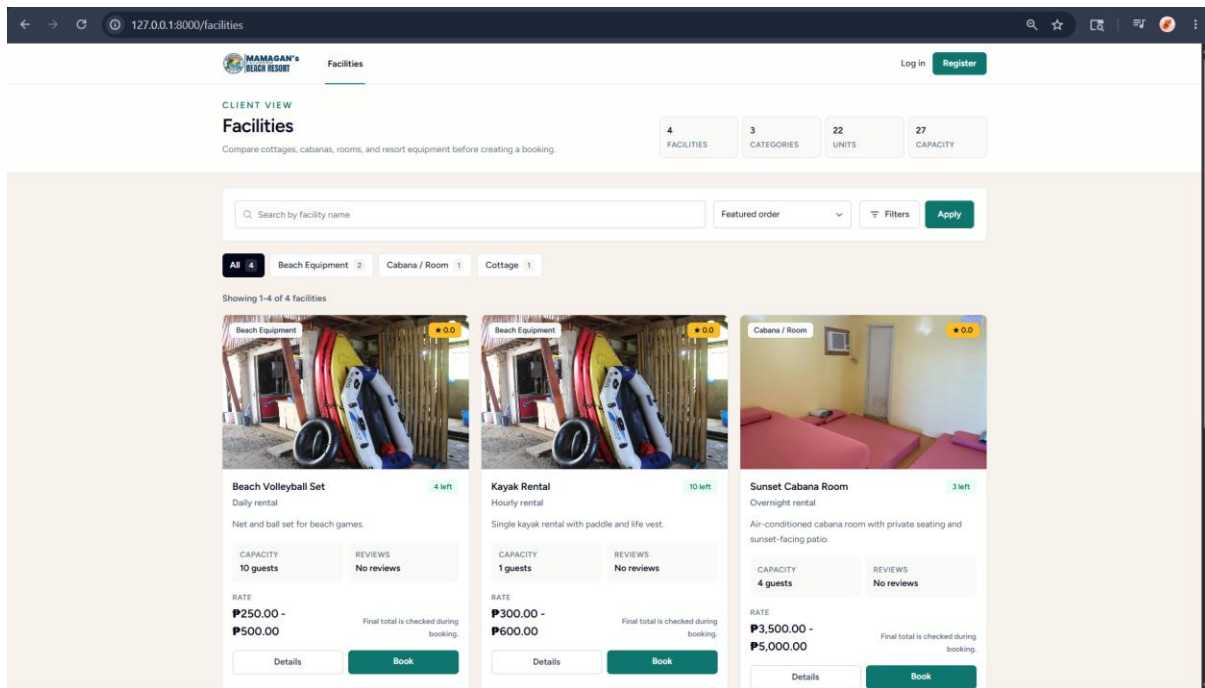


Figure 8: *Facility Browsing Page*

The facilities page allows guests to view available resort facilities such as cottages, cabanas, day beds, jet skis, and island tours. This page helps guests compare facilities before creating a booking.

The screenshot shows the 'Kayak Rental' facility details page on the Mamagan's Beach Resort website. The page includes a header with the resort logo, navigation links, and user options. The main content area features a large image of kayaks, a 'RATE RANGE' section showing prices from ₱300.00 to ₱600.00, and a 'Book This Facility' button. Below the image, there are tabs for 'CAPACITY', 'INVENTORY', 'RATING', and 'REVIEWS'. The 'Overview' section describes the facility as 'Single kayak rental with paddle and life vest.' A 'GOOD TO KNOW' box lists important information for guests.

Facilities

Kayak Rental
Hourly rental · 1 guest capacity · 10 available

RATE RANGE
₱300.00 - ₱600.00
The booking form checks the selected date, time, quantity, and any promo code before payment.

Facility type: Beach Equipment

Rental: Hourly

Capacity: 1 guests

Availability: 10 units

Book This Facility

CAPACITY
1 guests

INVENTORY
10 units

RATING
★ 0.0

REVIEWS
0

Overview
Single kayak rental with paddle and life vest.

GOOD TO KNOW

- Bookable online while inventory is available.
- Final totals can change with seasonal rates and promos.
- Approved bookings generate a QR ticket for check-in.

Figure 9: Facility Details Interface

The facility details page provides complete information about a selected resort facility. Guests can review the facility description, rate, capacity, and availability before proceeding to the booking form.

The screenshot shows the 'Sunset Cabana Room' booking form on the Mamagan's Beach Resort website. The form is divided into several sections: 'Schedule' for selecting the booking date, type, start time, and end time; 'Guests and Quantity' for specifying the number of units and guests; 'Optional Details' for entering a promo code and notes; and a 'Booking Summary' sidebar on the right showing the facility name, capacity, inventory, and rate range. A 'Create Booking' button is at the bottom right.

CREATE BOOKING
Sunset Cabana Room
Cabana / Room · Overnight · 3 available

Schedule
Choose the date and type of reservation.

Booking Date: 05/21/2025

Booking Type: Day use

Start Time: 07:29 AM

End Time: 05:25 PM

Guests and Quantity
Tell the resort how many units and guests to expect.

Quantity: 1

Guest Count: 1

Optional Details

Promo Code: OPTIONAL

Notes: Special requests or arrival notes

Booking Summary

Facility: Sunset Cabana Room

Capacity: 4

Inventory: 3

Rate: ₱3,500.00 - ₱5,000.00

Final amount is calculated after seasonal rate and promo checks. PayMongo checkout starts after this form is submitted.

Create Booking

Figure 10: Online Booking Form

The booking form allows guests to enter their reservation details, including the selected date, time, number of guests, and quantity. The system calculates the total amount before the guest confirms the booking.

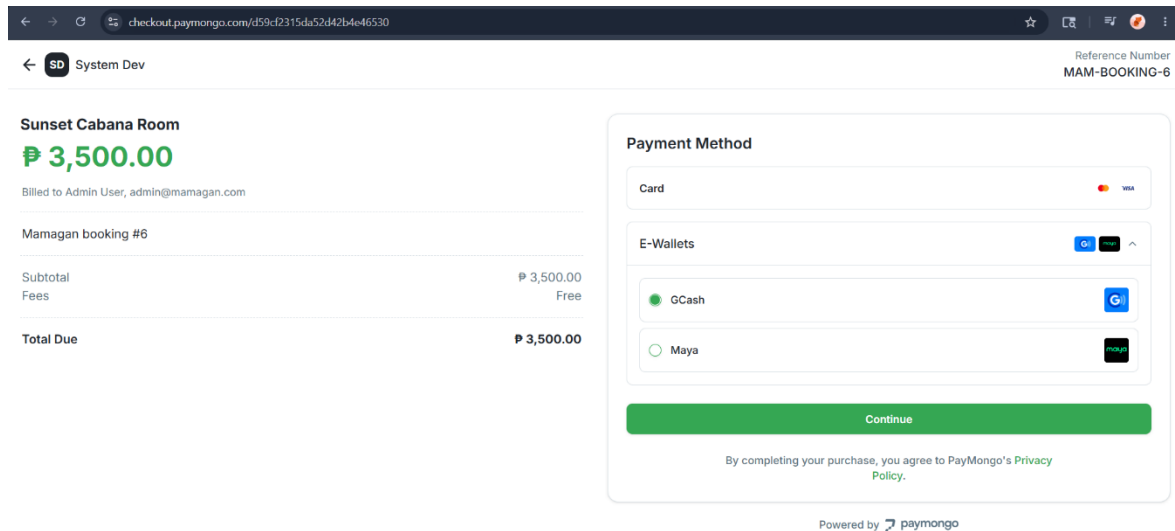


Figure 11: PayMongo Online Payment Interface

The payment page allows guests to complete their booking payment through PayMongo. The system supports online payment methods such as GCash, Maya, and card payments.

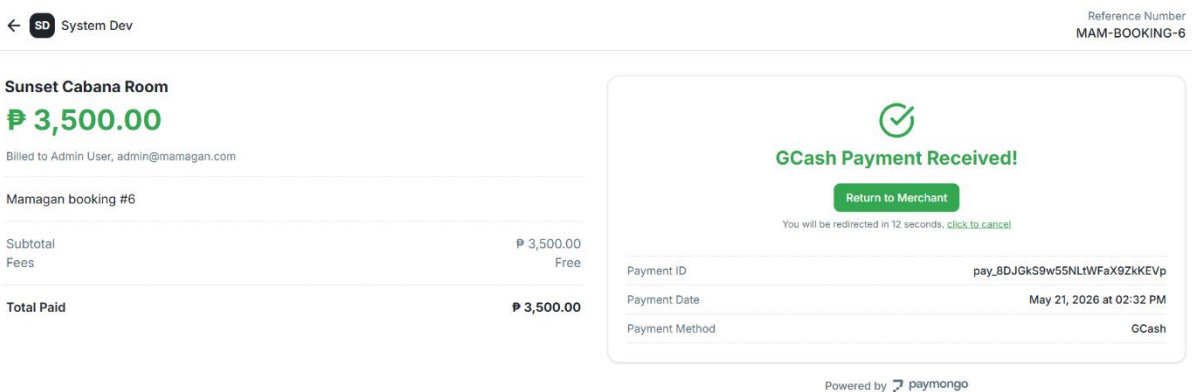


Figure 12: Booking Confirmation Page

The booking confirmation page informs the guest that the reservation has been successfully created and paid. It displays important booking details and confirms the status of the transaction.

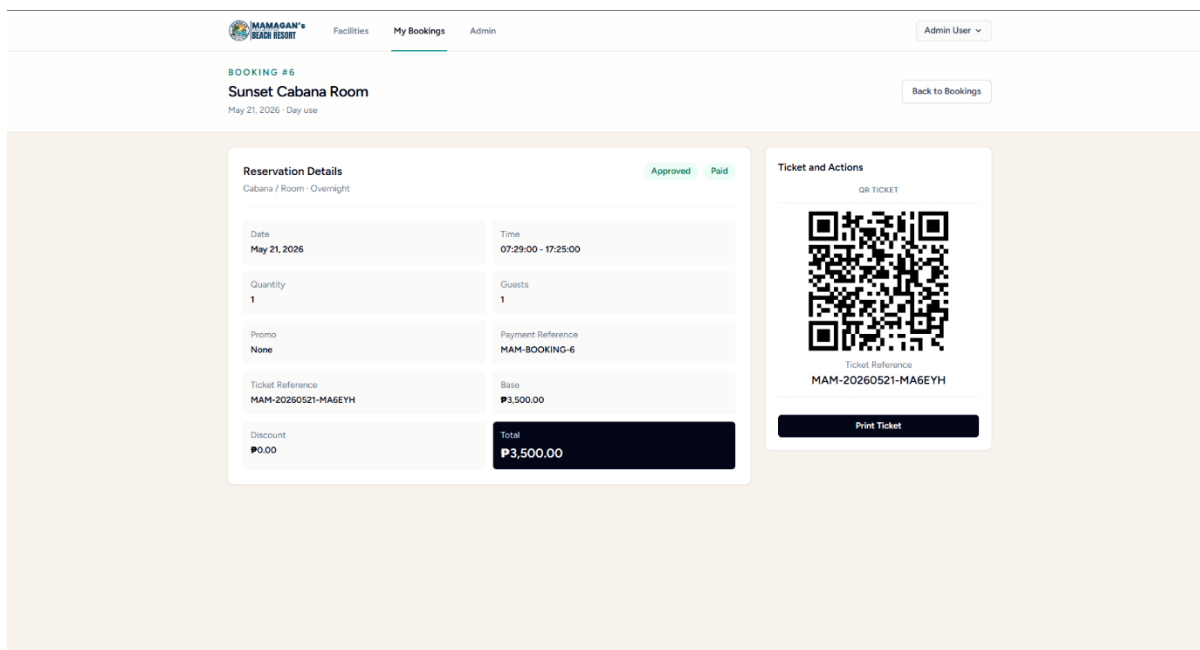


Figure 13: Digital QR Ticket Interface

The QR ticket page displays the digital ticket generated after successful payment. The guest can present this QR ticket upon arrival at the resort for faster booking verification and check-in.

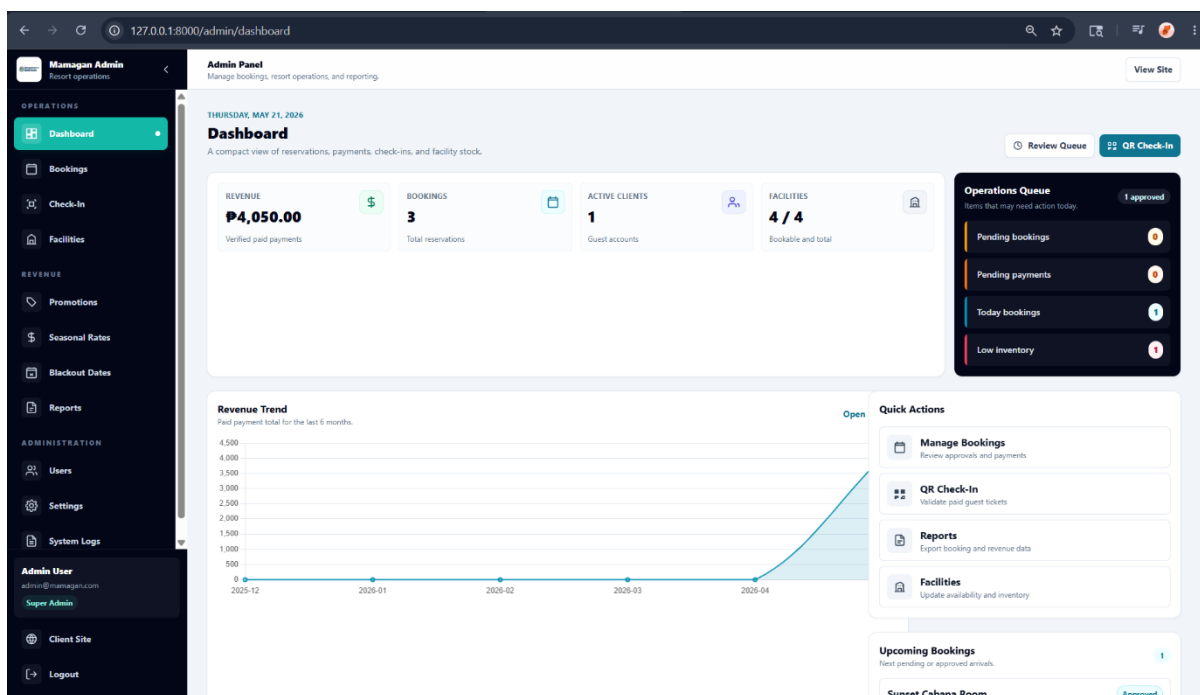


Figure 14: Admin Dashboard

The admin dashboard provides an overview of the resort's booking activities. It helps staff monitor reservations, payments, revenue, and other important system records.

DATA STRUCTURE

The database used for the Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration is named mamagan_resort. It is managed through MySQL/phpMyAdmin and contains tables for users, facilities, bookings, payments, QR tickets, promotions, reviews, system logs, and Laravel support functions.

Table Name	Description
users	Stores guest, staff, and admin account information
facilities	Stores resort facilities and services available for booking
bookings	Stores guest booking and reservation records
payments	Stores payment transaction details and PayMongo references
tickets	Stores QR ticket details generated after successful payment
promotions	Stores promo codes, discounts, and promotional offers
blackout_dates	Stores dates when facilities are unavailable
seasonal_rates	Stores special rates for selected seasons or date ranges
facility_reviews	Stores guest ratings and comments for facilities
facility_review_media	Stores uploaded images or media attached to reviews
email_verification_codes	Stores verification codes for user email verification
system_logs	Stores activity logs performed by users in the system
app_settings	Stores general resort and system settings
cottages	Stores cottage information, if used by the system
reservations	Stores cottage reservation records, if used by the system
sessions	Stores user session data
password_reset_tokens	Stores password reset tokens
migrations	Stores Laravel migration records

cache	Stores cached system data
cache_locks	Stores cache lock records
jobs	Stores queued jobs
job_batches	Stores grouped queued jobs
failed_jobs	Stores failed background jobs

Table 1: *List of Table*

Field Name	Description
id	Primary key of the user
name	Full name of the user
email	Email address of the user
phone	Contact number of the user
email_verified_at	Date and time when the email was verified
password	Encrypted password of the user
role	Defines the user role, such as guest, staff, or super admin
status	Indicates if the account is active or inactive
social_id	Stores social login ID, if used
social_type	Stores the type of social login, such as Google
remember_token	Token used for remember-me login
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 2: *User Table*

Field Name	Description
id	Primary key of the facility
name	Name of the facility
category	Category of the facility, such as cottage, cabana, or beach equipment
description	Details or description of the facility
image	Image path of the facility
price_min	Minimum price of the facility
price_max	Maximum price of the facility
capacity	Maximum number of guests allowed
inventory_count	Number of available facility units
rental_type	Type of rental, such as daily, hourly, or overnight
is_active	Indicates if the facility is active
is_bookable	Indicates if the facility can be booked
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 3: *Facilities Table*

Field Name	Description
id	Primary key of the booking
user_id	Foreign key connected to the users table
facility_id	Foreign key connected to the facilities table
promotion_id	Foreign key connected to the promotions table
booking_date	Date selected for the reservation
start_time	Starting time of the booking
end_time	Ending time of the booking
quantity	Number of facility units booked
guest_count	Number of guests included in the booking
booking_type	Type of booking, such as day use
base_amount	Original amount before discount
discount_amount	Discount amount applied to the booking
total_amount	Final amount to be paid
booking_status	Status of the booking, such as pending, confirmed, cancelled, or checked-in
payment_status	Status of payment, such as pending or paid
notes	Additional notes for the booking
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 4: *Bookings Table*

Field Name	Description
id	Primary key of the payment
booking_id	Foreign key connected to the bookings table
method	Payment method used, such as GCash
amount	Amount paid by the guest
status	Payment status, such as pending or paid
reference_number	Payment or booking reference number
paymongo_checkout_id	PayMongo checkout session ID
paymongo_payment_id	PayMongo payment transaction ID
checkout_url	PayMongo checkout link
provider_payload	Response data from PayMongo
proof_path	Path of uploaded proof of payment, if applicable
verified_at	Date and time when payment was verified
verified_by	User ID of staff or admin who verified the payment
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 5: *Payments Table*

Field Name	Description
id	Primary key of the ticket
booking_id	Foreign key connected to the bookings table
reference_number	Unique ticket reference number
qr_payload	Data stored in the QR ticket
issued_at	Date and time when the ticket was issued
checked_in_at	Date and time when the guest checked in
checked_in_by	User ID of staff or admin who confirmed check-in
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 6: *Tickets Table*

Field Name	Description
id	Primary key of the promotion
name	Name of the promotion
code	Promo code used by guests
discount_type	Type of discount, such as percentage or fixed amount
discount_value	Value of the discount
minimum_amount	Minimum amount required to apply the promo
starts_at	Start date of the promotion
ends_at	End date of the promotion
usage_limit	Maximum number of times the promo can be used
used_count	Number of times the promo has been used
facility_id	Foreign key connected to the facilities table

is_active	Indicates if the promotion is active
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 7: *Promotion Table*

Field Name	Description
id	Primary key of the blackout date
facility_id	Foreign key connected to the facilities table
title	Title or name of the blocked schedule
starts_at	Start date of unavailability
ends_at	End date of unavailability
reason	Reason for blocking the date
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 8: *Blackout Date Table*

Field Name	Description
id	Primary key of the seasonal rate
facility_id	Foreign key connected to the facilities table
name	Name of the seasonal rate
starts_at	Start date of the seasonal rate
ends_at	End date of the seasonal rate
price	Special price during the selected season
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 9: *Seasonal Rate Table*

Field Name	Description
id	Primary key of the review
user_id	Foreign key connected to the users table
facility_id	Foreign key connected to the facilities table
booking_id	Foreign key connected to the bookings table
rating	Rating given by the guest
comment	Guest comment or feedback
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 10: *Facility Review Table*

Field Name	Description
id	Primary key of the media record
facility_review_id	Foreign key connected to the facility_reviews table
path	File path of the uploaded media
media_type	Type of media, such as image
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 11: *Facility Review Media Table*

Field Name	Description
id	Primary key of the verification record
user_id	Foreign key connected to the users table
code	Verification code
expires_at	Expiration date and time of the code
verified_at	Date and time when the code was verified
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 12: *Email Verification Code Table*

Field Name	Description
id	Primary key of the log
user_id	Foreign key connected to the users table
action	Action performed by the user
target_type	Type of record affected
target_id	ID of the affected record
properties	Additional log details
ip_address	IP address of the user
created_at	Date and time the record was created
updated_at	Date and time the record was updated

Table 13: *System Logs Table*

IMPLEMENTATION AND TESTING

Deployment Process

The development of the Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration started with the identification of the problems in the existing manual booking process of the resort. The main issues observed were unorganized reservation records, possible overbooking, delayed payment confirmation, and difficulty in managing facility availability. These problems became the basis for designing a web-based system that could improve the booking and payment process.

After identifying the system requirements, the database structure was designed using MySQL and managed through phpMyAdmin. The database included tables for users, facilities, bookings, payments, tickets, promotions, blackout dates, seasonal rates, reviews, and system logs. These tables were created to organize the important records needed by the system.

The system was then developed using PHP and Laravel. Laravel was used to organize the backend structure, including routes, controllers, models, authentication, and database operations. The guest side of the system was created to allow users to register, log in, browse facilities, create bookings, and proceed with online payment. The admin side was developed to allow resort staff to manage facilities, bookings, payments, promotions, blocked dates, and reports.

The payment feature was implemented using PayMongo, which supports online payment methods such as GCash and other digital payment options. After a successful payment, the system generates a digital QR ticket that serves as proof of reservation. GitHub was also used during development for source code storage, backup, and version control.

Testing Procedures Performed

Testing Procedure	Description
User Registration Testing	Checked if new guests could successfully create an account.
Login Testing	Tested if guests, staff, and admins could log in using valid credentials.

Facility Display Testing	Checked if facility details such as name, price, capacity, and description were displayed correctly.
Booking Form Testing	Tested if guests could select a facility, date, time, quantity, and number of guests.
Availability Testing	Checked if the system could prevent bookings on unavailable or blocked dates.
Payment Testing	Tested if guests could proceed to PayMongo checkout and complete payment.
QR Ticket Testing	Checked if the system could generate a QR ticket after successful payment.
Admin Dashboard Testing	Tested if the admin could view bookings, payments, reports, and system records.
Facility Management Testing	Checked if admin users could add, edit, update, or deactivate facilities.
Promotion Testing	Tested if promo codes and discounts were applied correctly.
Report Testing	Checked if booking, revenue, and occupancy reports displayed correct data.
Database Testing	Verified if records were properly saved, updated, and retrieved from MySQL.

Table 14: *Testing Procedures*

Problem Encountered

During the development and testing of the system, several problems were encountered.

First, there were issues in connecting the Laravel system to the MySQL database. Some database credentials and environment settings needed to be corrected before the system could properly connect to phpMyAdmin/MySQL.

Second, some booking records did not save correctly during initial testing because of missing fields, incorrect form validation, or mismatched database column names. This affected the

booking process and required checking of the booking form, controller, and database table structure.

Third, payment integration with PayMongo required proper configuration of API keys, checkout sessions, and payment status updates. At first, some payment records remained pending because the system needed correct handling of PayMongo responses.

Fourth, QR ticket generation required proper linking between the booking, payment, and ticket records. The QR ticket should only be generated after a successful payment, so the system logic needed to verify payment status before creating the ticket.

Lastly, some admin functions needed adjustments, especially in managing facilities, blocked dates, promotions, and reports. These features required proper validation to prevent incorrect or incomplete data from being saved.

Solution Applied

To solve the database connection issue, the .env configuration file was checked and updated with the correct database name, username, password, and server settings. The database tables were also reviewed in phpMyAdmin to make sure they matched the Laravel models and migrations.

For booking problems, form validation was improved to ensure that all required fields were completed before submitting a booking. The booking controller and database columns were also checked to make sure that the data entered by the guest was saved correctly.

For payment issues, the PayMongo API configuration was reviewed and tested. The system was adjusted to properly store payment references, checkout IDs, payment status, and transaction details in the payments table. This helped improve payment monitoring and confirmation.

For QR ticket generation, the system was configured to generate a ticket only after the payment status was successfully updated. The QR ticket was connected to the booking record so that staff could verify the reservation during guest check-in.

For admin management issues, validation and update functions were improved. Facility records, promotions, seasonal rates, and blackout dates were tested repeatedly to make sure that admins could properly add, edit, and update system data.

CONCLUSION

The Mamagan Fun & Adventure Beach Resort Online Booking and Payment System with GCash Integration successfully achieved its purpose of providing a web-based platform for managing resort reservations, online payments, facility availability, and booking confirmation. The project was able to develop important features such as guest registration and login, facility browsing, online booking, PayMongo payment integration, QR ticket generation, admin dashboard, facility management, booking monitoring, and basic reporting.

The system benefits both guests and resort staff. For guests, it provides a more convenient way to view facilities, check available schedules, make reservations, and pay online without needing to visit the resort personally. For resort staff, it helps organize booking records, monitor payments, manage facilities, prevent overbooking, and confirm guest arrivals using QR tickets. This makes the reservation process faster, more accurate, and more efficient compared to manual booking methods.

During the development of the project, the developers gained knowledge and experience in using PHP, Laravel, MySQL, PayMongo, and GitHub. The project also helped improve skills in database design, backend development, payment gateway integration, system testing, troubleshooting, and web application management. Overall, the project served as a valuable learning experience in developing a real-world system that addresses actual problems in resort booking and management.

RECOMMENDATIONS

Although the system already provides the main functions needed for online booking and payment, several improvements are recommended for future development.

First, the system may be improved by adding a mobile application version for Android and iOS users. This would make the booking process more accessible and convenient for guests who prefer using mobile apps.

Second, the system may include SMS notification or email notification features to automatically inform guests about booking confirmation, payment status, QR ticket details, reminders, and cancellation updates.

Third, the system may be enhanced by adding an automatic refund processing feature connected to the payment gateway. This would make cancellation and refund handling more convenient for both guests and resort staff.

Fourth, a chatbot or live chat support feature may be added to assist guests with common questions about facilities, rates, availability, and booking procedures.

Fifth, the admin dashboard may be improved with more advanced analytics and reporting features, such as monthly revenue reports, most booked facilities, guest statistics, and occupancy trends.

Sixth, the system may include walk-in booking management so that staff can record onsite reservations directly in the system and keep all booking records centralized.

Lastly, future developers may improve the system's security by adding features such as two-factor authentication, stronger role-based access control, regular database backup, and improved activity monitoring.

These recommendations can help make the system more reliable, user-friendly, secure, and useful for the future operations of Mamagan Fun & Adventure Beach Resort.